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15

Assignment 2
Multiple Linear Regression

A.

A wind energy company wants to predict the Power Output (kW) of a wind turbine using the following parameters:
- wind speed (m/s)
- Blade angle (°)
- Rotor Speed (RPM)

Dataset

Wind Speed (m/s)	Blade Angle (°)	Rotor Speed (RPM)	Power Output (kW)
5	11	125	48
7	9	155	72
9	15	175	101
11	13	220	142
13	18	235	176

Normal equation

$$\theta = (X^T X)^{-1} X^T Y$$

where

$X \rightarrow$ Matrix of input features

$Y \rightarrow$ Target variable (Power Output)

$\theta \rightarrow$ Regression coefficients

Matrix formation

$$X = \begin{bmatrix} 1 & 5 & 11 & 125 \\ 1 & 7 & 9 & 155 \\ 1 & 9 & 15 & 175 \\ 1 & 11 & 13 & 220 \\ 1 & 13 & 18 & 235 \end{bmatrix} \qquad Y = \begin{bmatrix} 48 \\ 72 \\ 101 \\ 142 \\ 176 \end{bmatrix}$$

$\therefore$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 11 & 9 & 15 & 13 & 18 \\ 125 & 155 & 175 & 220 & 235 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 11 & 9 & 15 & 13 & 18 \\ 125 & 155 & 175 & 220 & 235 \end{bmatrix} \begin{bmatrix} 1 & 5 & 11 & 125 \\ 1 & 7 & 9 & 155 \\ 1 & 9 & 15 & 175 \\ 1 & 11 & 13 & 220 \\ 1 & 13 & 18 & 235 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 5 & 45 & 66 & 910 \\ 45 & 445 & 630 & 8760 \\ 66 & 630 & 920 & 12485 \\ 910 & 8760 & 12485 & 173900 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{|X^T X|} \operatorname{adj}(X^T X)$$

$$|X^T X| = \begin{vmatrix} 5 & 45 & 66 & 910 \\ 45 & 445 & 630 & 8760 \\ 66 & 630 & 920 & 12485 \\ 910 & 8760 & 12485 & 173900 \end{vmatrix} = 196600$$

$$|X^T X| = 196600$$

Using Calculator

$$(X^TX)^{-1} = \begin{bmatrix} 76.9017 & 18.2148 & -3.0043 & -1.1043 \\ 18.2148 & 4.5850 & -0.7240 & -0.2744 \\ -3.0043 & -0.7240 & 0.1600 & 0.0407 \\ -1.1042 & -0.2744 & 0.0406 & 0.0167 \end{bmatrix}$$

$$X^TY = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 11 & 9 & 15 & 13 & 18 \\ 125 & 155 & 175 & 220 & 235 \end{bmatrix} \begin{bmatrix} 48 \\ 72 \\ 101 \\ 142 \\ 178 \end{bmatrix}$$

$$X^TY = \begin{bmatrix} 539 \\ 5503 \\ 7705 \\ 107435 \end{bmatrix}$$

$$\theta = (X^TX)^{-1} X^TY$$

$$\theta = \begin{bmatrix} 76.9017 & 18.2148 & -3.0043 & -1.1043 \\ 18.2148 & 4.5850 & -0.7240 & -0.2744 \\ -3.0043 & -0.7240 & 0.1600 & 0.0407 \\ -1.1042 & -0.2744 & 0.0406 & 0.0167 \end{bmatrix} \begin{bmatrix} 539 \\ 5503 \\ 7705 \\ 107435 \end{bmatrix}$$

$$\theta = \begin{bmatrix} -99.9112 \\ 1.2174 \\ 2.4186 \\ 0.9058 \end{bmatrix} = \begin{bmatrix} c \\ m_1 \\ m_2 \\ m_3 \end{bmatrix}$$

$c = -99.9112$

$m_1 = 1.2174$

$m_2 = 2.4186$

$m_3 = \cancel{0.90158}$ 0.9058

Page No.

Date / /

ML model equation

$Y = m_1 x_1 + m_2 x_2 + m_3 x_3 + c$

For wind speed $(x_1) = 9$

Blade angle $(x_2) = 11$

rotor speed $(x_3) = 100$

Power output $(Y) = (1.2174 \times 9) + (2.4166 \times 11) + (0.9058 \times 100) + (-99.9112)$

$Y = \cancel{22}\ 28.208$

$\therefore$ the required power output is 28.208 kW

```python
import pandas as pd
from sklearn.linear_model import LinearRegression
data = pd.read_csv("power_data.csv")
x = data[["Wind_Speed", "Blade_Angle", "Rotor_Speed"]]
y = data[["Power_Output"]]

model = LinearRegression()
model.fit(x,y)
print("Coeff: ",model.coef_)
print("Intercept: ",model.intercept_)

data_1 = 9,11,100
power_output_1 = model.predict([data_1])
print("The power output for (Wind_Speed=9, Blade_Angle=11, Rotor_Speed=100) is: ", power_output_1)

Wind_Speed= int(input("Enter the Wind_Speed : "))
Blade_Angle= int(input("Enter the Blade_Angle : "))
Rotor_Speed= int(input("Enter the Rotor_Speed : "))
power_output = model.predict([[Wind_Speed,Blade_Angle,Rotor_Speed]])
print("The power output is: ", power_output)
```

```
PS D:\coding> & C:\Users\alphi\AppData\Local\Programs\Python\Python314\python.exe d:/coding/Power_output.py
Coeff:  [[1.21744659 2.41658189 0.90579858]]
Intercept:  [-99.9112411]
C:\Users\alphi\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
The power output for (Wind_Speed=9, Blade_Angle=11, Rotor_Speed=100) is:  [[28.20803662]]
Enter the Wind_Speed : 12
Enter the Blade_Angle : 19
Enter the Rotor_Speed : 165
C:\Users\alphi\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
The power output is:  [[110.06993896]]
PS D:\coding>
```